Lab 3: Component Modelling & Architectural Pattern Selection

Name:-Pavithra G C

SRN:-PES2UG24CS346   SECTION:-5F

Scenario Details

Scenario: Dropshipping Inventory & Order Orchestrator

Primary Domain: Retail, E-Commerce & Finance

Target Actors:

1. E-Store Owner
2. Supplier Partner

1. Functional Requirements

The Dropshipping Inventory & Order Orchestrator should provide the following functions:

1. **Order Management** – Manage and track customer orders.
2. **Inventory Management** – Check product availability and update stock.
3. **Payment Processing** – Process and verify payments.
4. **Supplier Management** – Communicate with supplier partners and manage supplier information.
5. **Order Fulfillment** – Send orders to suppliers and receive fulfillment updates.
6. **Inventory Synchronization** – Synchronize inventory information with suppliers.
7. **Order Notifications** – Send order and status updates to the E-Store Owner.

8. **Order Status Tracking** – Track the progress of an order from placement to fulfillment.

2. Non-Functional Requirements

The Dropshipping Inventory & Order Orchestrator should satisfy the following non-functional requirements:

1. **Performance** – The system should process orders and inventory requests quickly.
2. **Scalability** – The system should support an increasing number of orders and supplier partners.
3. **Security** – Payment, order, and business information should be protected from unauthorized access.
4. **Availability** – The system should remain available even if one service experiences a failure.
5. **Reliability** – Order and inventory information should be accurate and consistent.
6. **Maintainability** – Individual services should be easy to update and maintain.

3. Main Challenges

The Dropshipping Inventory & Order Orchestrator faces the following challenges:

1. **Inventory Synchronization** – Supplier inventory may change frequently, so the system must keep inventory information updated and accurate.
2. **High Order Volume** – The system should handle many orders simultaneously, especially during peak sales periods.
3. **Payment Security** – Payment-related information must be protected from unauthorized access.
4. **Supplier Communication** – The system must reliably communicate with different supplier partners for inventory and order fulfillment.
5. **Service Failure** – Failure of one service should not unnecessarily stop the complete order processing system.

4.Selected Architecture: Microservices Architecture

For the Dropshipping Inventory & Order Orchestrator, we select **Microservices Architecture**.

Microservices is suitable because the system contains different functions such as order management, payment processing, inventory management, supplier management, and notifications. Each function can work as an independent service.

Components

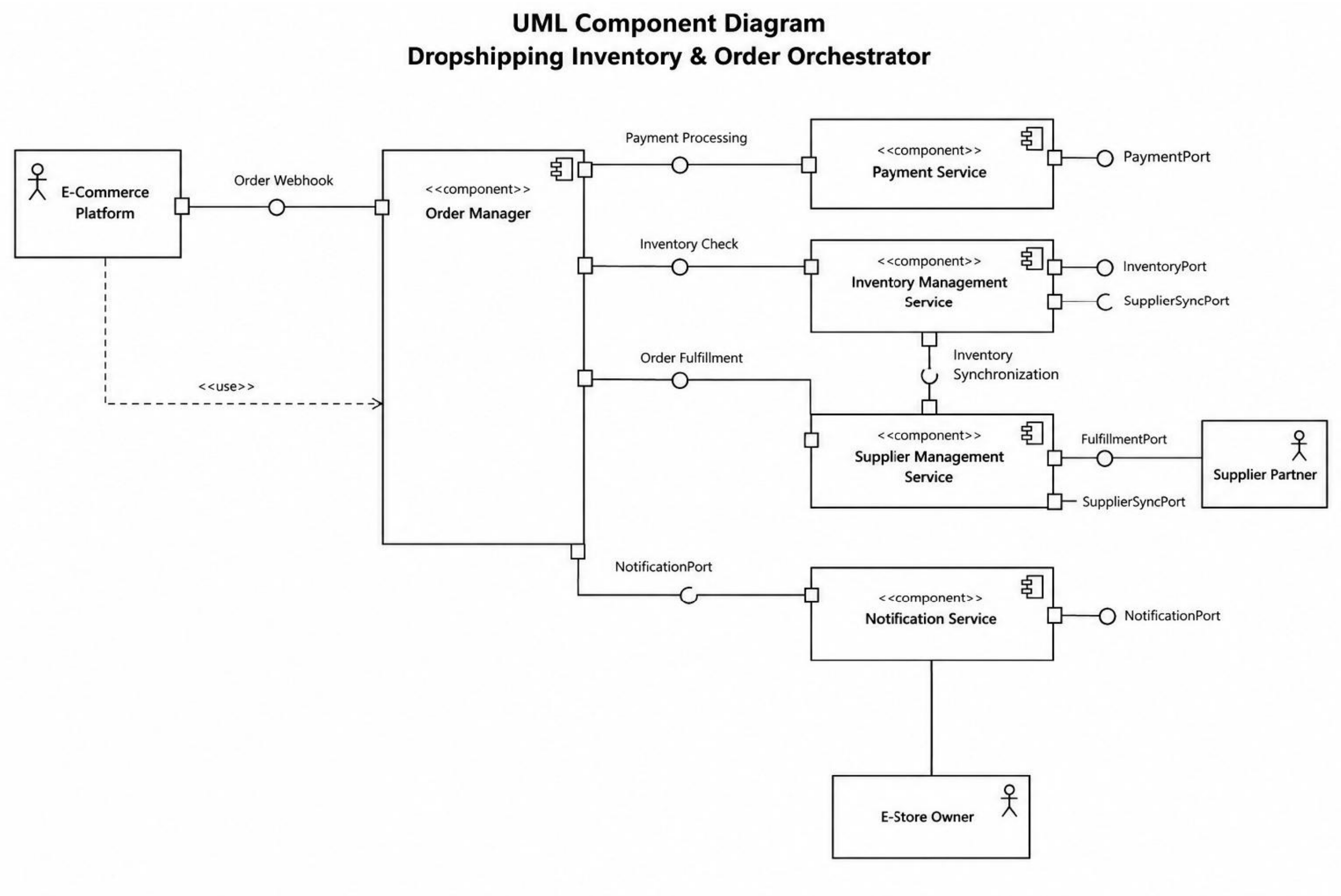
The following five components are used:

1. **Order Manager Component** – Manages and tracks customer orders.
2. **Payment Service Component** – Processes and verifies payments.
3. **Inventory Management Component** – Checks product availability and updates stock.
4. **Supplier Management Component** – Communicates with supplier partners and manages supplier information.
5. **Notification Service Component** – Sends order and status notifications to the E-Store Owner.

Why Microservices?

- Each component can be developed and maintained independently.
- Individual services can be scaled according to demand.
- Failure of one service does not necessarily stop the entire system.
- It is suitable for handling high order volumes and multiple suppliers.

Component Diagraml-



Architectural Justification

Architecture Selected: Microservices Architecture

We selected **Microservices Architecture** for the Dropshipping Inventory & Order Orchestrator because the system contains different functions such as order management, payment processing, inventory management, supplier management, and notifications.

Reason 1 – Independent Scaling

Each service can be scaled independently based on its workload. For example, during peak sales, the Order Manager and Inventory Management services can be scaled without scaling the other services.

Reason 2 – Fault Isolation

Microservices provide better fault isolation. If one service fails, such as the Notification Service, other important services such as Order Manager and Payment Service can continue working.

Security Advantage

The Payment Service can be isolated from other services and protected using authentication and authorization. This helps protect sensitive payment and order information.

Performance Advantage

Frequently used services such as Order Manager and Inventory Management can be independently scaled. This helps the system handle a large number of orders and inventory requests efficiently.

Conclusion

Microservices Architecture is suitable for the Dropshipping Inventory & Order Orchestrator because it provides scalability, fault isolation, security, and better performance for a system with multiple independent functions.